STATISTICAL INFERENCE ASSIGNMENT

Statistical Investigation of a Country Using World Bank Data

Individual Assignment • World Development Indicators • 2010–2024

1. Assignment Question

1. How did the economic, social, infrastructure, technological and environmental indicators of a particular country evolve between 2010 and 2024?
2. What statistical patterns and relationships can be identified in the data, and what can and cannot be concluded from those statistics?

You will investigate one country using a common set of World Bank indicators. Every student will follow the same analytical structure so that the reports are comparable. The purpose is not to write a general country profile.

The purpose is to demonstrate the complete statistical workflow: obtaining data, documenting it, cleaning it, describing it, presenting it, measuring variation, making statistical inferences, testing relationships, and interpreting the evidence.

Your report must show the data and the analysis process, not only the final numerical results. A reader should be able to trace each major result back to the country dataset you submit.

2. What You Are Required to Submit

- Approximately 2,500–3,000 words, excluding tables, figures, references and appendix.
- Excel (.xlsx), CSV (.csv), or Stata (.dta) containing the World Bank observations used in the report.
- Your Stata/R/Python/Excel calculations or equivalent reproducible work. If you work entirely in Excel, submit the workbook containing formulas/calculations.
- Variable definitions, data-quality notes, key calculations, and any additional output needed to verify the report.

3. Country Allocation and Country Switching

Each student will work on one country. No two students should submit from the same country unless explicitly authorized by the instructor.

1. Select a country.
2. Download/check all nine required indicators for 2010–2024.
3. If the country has inadequate data coverage, you may switch to another country. Check the data first rather than discovering the problem after completing the analysis.
4. Prefer countries with at least 10 valid annual observations for most indicators and enough paired observations for the required correlation analyses. If several indicators are largely unavailable, choose another country.

Do not fill missing World Bank observations by interpolation, extrapolation, carrying values forward/backward, or personal estimates. Missing observations must remain missing and must be reported.

4. Data Source and Required Indicators

Use the World Bank World Development Indicators (WDI). Download the observations for the selected country for 2010–2024. Use the indicator codes below so that every student works with the same definitions.

No.	Indicator	World Bank code	Unit	Primary SDG context	Required use
1	GDP per capita	NY.GDP.PCAP.KD	constant 2015 US$ per person	SDG 8	Descriptive, trend, dispersion, correlations
2	GDP growth	NY.GDP.MKTP.KD.ZG	annual %	SDG 8	Descriptive, trend, dispersion
3	Life expectancy at birth	SP.DYN.LE00.IN	years	SDG 3	Descriptive, trend, dispersion, correlation
4	People using at least basic drinking-water services	SH.H2O.BASW.ZS	% of population	SDG 6	Descriptive, trend, dispersion
5	School enrollment, secondary (% gross)	SE.SEC.ENRR	% gross	SDG 4	Descriptive, trend, dispersion
6	Access to electricity	EG.ELC.ACCS.ZS	% of population	SDG 7	Descriptive, trend, dispersion, correlation
7	Individuals using the Internet	IT.NET.USER.ZS	% of population	SDG 9	Descriptive, trend, dispersion
8	CO2 emissions per capita	EN.ATM.CO2E.PC	metric tons per capita	SDG 13	Descriptive, trend, dispersion, correlation
9	Forest area	AG.LND.FRST.ZS	% of land area	SDG 15	Descriptive, trend, dispersion

Dataset identification variables: Year, Country, Country Code, and the nine indicators above.

Important: The SDG column provides substantive context. Some indicators can relate to more than one SDG; the table assigns a primary SDG context for this assignment. The exercise is not an SDG ranking or SDG performance score.

5. Step-by-Step Workflow

Step 1 — Formulate the statistical investigation

Begin with the common research question above. Then write 3–5 specific sub-questions that can actually be answered using the assigned data.

- How did the country's GDP per capita and GDP growth change over 2010–2024?
- Which indicators show the greatest and lowest variability?
- Are GDP per capita and life expectancy statistically associated in this country's annual data?
- Is GDP per capita associated with electricity access or CO2 emissions per capita?
- What unusual observations or data-quality issues need to be documented?

Important: Do not introduce a new research design that requires variables outside the common indicator set. Your sub-questions should support the statistical methods required below.

Step 2 — Download and document the data

1. Download the nine indicators for your country from the World Bank WDI.
2. Keep the original downloaded values. Do not overwrite the raw data.
3. Create a clean analysis dataset with one row per year and one column per indicator.
4. Restrict the analysis period to 2010–2024.
5. Record the indicator code, indicator name, unit, source, download date, and definition.
6. Record the number of potential observations (15 years), actual non-missing observations, and missing observations for every indicator.
7. Check for duplicates, impossible values, inconsistent units, and unexpected changes.
8. If you switch countries because of poor coverage, document the screening evidence in the appendix.

Step 3 — Describe the data before analysing it

Your report must explain what each variable measures and how it is measured. Do not simply reproduce the World Bank variable name.

1. Identify each variable as quantitative and, where appropriate, continuous/discrete.
2. State the unit and interpretation of the unit.
3. Explain the time dimension: annual observations from 2010 to 2024.
4. Distinguish the observed country-year data from a broader population. Do not automatically treat 15 annual observations as a random sample from all years.
5. Discuss missing observations and their implications.

Step 4 — Produce descriptive statistics

For every one of the nine indicators, calculate:

- Number of non-missing observations (N)
- Mean
- Median
- Minimum
- Maximum
- Range
- Variance
- Standard deviation
- First quartile (Q1)
- Third quartile (Q3)
- Interquartile range (IQR)

Present these in one consolidated descriptive-statistics table. Then select three indicators for detailed interpretation.

- Compare the mean and median and comment on possible skewness.
- Explain what the minimum and maximum imply in the country's observed period.
- Explain whether the mean is a useful summary for each selected indicator.

- Do not interpret a large numerical variance as automatically meaning 'more unstable' when variables use different units.

Step 5 — Present the data graphically

Your report must contain:

- One time-series graph for each of the nine indicators, with Year on the horizontal axis.
- Two histograms for selected indicators, with a short justification for the selection.
- Two boxplots for selected indicators, with interpretation of median, quartiles, spread, and potential outliers.
- Scatterplots for each of the three required correlation pairs

Important: Every figure must have a number, an informative title, labelled axes, units where relevant, and a source note. A graph without interpretation does not count as analysis.

Step 6 — Analyse variance and heterogeneity

This section is a central part of the assignment. For four indicators, calculate and interpret:

- Variance
- Standard deviation
- Coefficient of variation (CV), where meaningful
- Range
- IQR

Then answer the following:

- Which indicators exhibit the greatest and lowest variability?
- Does the ordering change when you use variance, standard deviation, CV, range, and IQR?
- Why can raw variance be misleading when indicators have different units and scales?
- Which measure is most informative for each selected indicator, and why?

Important: Do not compare variances mechanically across GDP per capita, percentages, years, and tons per capita. Discuss scale and units before making cross-variable comparisons.

Step 7 — Construct and interpret confidence intervals

Construct two 95% confidence intervals using methods covered in the course by the submission deadline. Choose the parameters and methods that are appropriate for the available data.

- State the parameter being estimated.
- State the estimator.
- Show the relevant standard error or calculation.
- State the critical value/method used.
- Report the confidence interval numerically.
- Interpret the interval in words.

- State the assumptions required by the method.
- Explain any limitations created by the small number of annual observations or time-series structure.

Important: Do not write 'there is a 95% probability that the true parameter lies in this interval.' Use the frequentist interpretation taught in class.

Step 8 — Conduct two hypothesis tests

Conduct two hypothesis tests using statistical methods actually covered in the course by the assignment deadline. The exact test should match the question, variable type, and assumptions.

- State the research question.
- State H0 and H1 clearly.
- Choose and state the significance level (use $\alpha = 0.05$ unless there is a justified reason to use another level).
- Report the test statistic and p-value or critical-value comparison.
- State the decision regarding H0.
- Translate the statistical decision into substantive language about the country.
- Discuss the assumptions and limitations of the test.

Important: A small p-value is evidence against H0 under the specified assumptions; it is not by itself a measure of practical importance or causality.

Step 9 — Covariance and correlation

Calculate covariance and Pearson correlation for all three pairs:

- GDP per capita × Life expectancy
- GDP per capita × Access to electricity
- GDP per capita × CO2 emissions per capita
- Report the covariance and Pearson's correlation coefficient for each pair.
- Include the corresponding scatterplot.
- Describe the direction and strength of the observed association.
- If significance testing for correlation has been taught by the submission deadline, report the test and p-value.
- Use the number of paired non-missing observations (N) for each correlation.
- Explain why correlation does not establish causation.

Step 10 — Confounding and causal interpretation

Select one of the three correlations and identify at least one plausible third variable that could influence both variables. Explain the confounding problem using a simple causal diagram:

X ← Z → Y

- Define X, Y, and Z in your example.

- Explain why Z could be associated with both X and Y.
- Explain how omitting Z could create or distort the observed association.
- State clearly what your correlation analysis can show and what it cannot show.

Important: Do not perform regression, instrumental variables, propensity-score matching, or other advanced causal methods unless the instructor explicitly requires them. This assignment is designed to establish the statistical foundation before econometrics.

Step 11 — Investigate one unusual observation or data-quality feature

Choose one observation that deserves investigation: a sudden increase/decrease, unusual outlier, missing year, apparent break in a series, or other data-quality feature.

- Identify the year and variable.
- Show the value and the surrounding years.
- Explain why it is unusual.
- Check the World Bank metadata or another authoritative source where appropriate.
- State whether you believe it represents a genuine change, a measurement/data issue, a revision, or an unresolved feature.
- Do not delete or alter the observation simply because it looks unusual.

6. Required Report Structure

Report section	What it must contain
1. Introduction	Country, purpose, main research question, and 3–5 sub-questions.
2. Data and Measurement	Source, period, country, indicator definitions, units, data structure, missingness, and any country switch.
3. Descriptive Statistics	Consolidated table and interpretation of selected indicators.
4. Graphical Analysis	Time-series graphs, histograms, boxplots, and interpretation.
5. Variance and Heterogeneity	Variance, SD, CV, range, IQR and cross-measure comparison.
6. Confidence Intervals	Two 95% CIs with calculations, assumptions, and interpretation.
7. Hypothesis Testing	Two tests with H0/H1, test statistic, p-value/critical value, decision, interpretation, assumptions.
8. Covariance and Correlation	Three required pairs, covariance, Pearson r, scatterplots, and interpretation.
9. Confounding and Limits of Interpretation	One confounder example and explanation of correlation vs causation.
10. Conclusion	What the statistics collectively say about the country, plus limitations.
Appendix	Variable dictionary, data-quality check, unusual-observation investigation, key calculations, and additional output.

7. How to Report Statistical Results

Use the following sequence whenever you report a statistical result:

- State what was calculated or tested.
- Show the numerical result.
- Explain what the number means in context.
- State the relevant assumption or limitation.
- Avoid causal language unless a causal design justifies it.

Example of acceptable reporting: "The Pearson correlation between GDP per capita and life expectancy is r = __ based on __ paired annual observations. The positive value indicates that the two variables tended to move in the same direction during the observed period. This is an association within the country's annual data; it does not by itself establish that changes in GDP per capita caused changes in life expectancy."

Avoid: "GDP causes life expectancy to increase because the correlation is high."

8. Data and Calculation Requirements

- Keep a copy of the original World Bank download.
- Do not manually type numbers into the final report when the value can be linked to your dataset/calculation.
- Retain full precision in calculations; round only when presenting results.
- Use a consistent number of decimal places across comparable statistics.
- Label missing values explicitly.
- Do not silently drop observations. If an observation is excluded from a calculation because of missingness, report the resulting N.
- Make sure every table and figure can be reproduced from the submitted data/calculation file.

9. Suggested Variable Dictionary

Include a table like the following in your appendix and complete it with the exact metadata you downloaded.

Variable	World Bank code	Definition	Unit	N / Missing
GDP per capita	NY.GDP.PCAP.KD	Copy the World Bank definition used.	constant 2015 US$ per person	Report N and missing years.
GDP growth	NY.GDP.MKTP.KD.ZG	Copy the World Bank definition used.	annual %	Report N and missing years.
Life expectancy at birth	SP.DYN.LE00.IN	Copy the World Bank definition used.	years	Report N and missing years.
People using at least basic drinking-water services	SH.H2O.BASW.ZS	Copy the World Bank definition used.	% of population	Report N and missing years.
School enrollment, secondary (% gross)	SE.SEC.ENRR	Copy the World Bank definition used.	% gross	Report N and missing years.
Access to electricity	EG.ELC.ACCS.ZS	Copy the World Bank definition used.	% of population	Report N and missing years.
Individuals using the Internet	IT.NET.USER.ZS	Copy the World Bank definition used.	% of population	Report N and missing years.
CO2 emissions per capita	EN.ATM.CO2E.PC	Copy the World Bank definition used.	metric tons per capita	Report N and missing years.
Forest area	AG.LND.FRST.ZS	Copy the World Bank definition used.	% of land area	Report N and missing years.

10. Marking Scheme — 100 points

Component	Points
1. Introduction and research questions	5
2. Data construction and documentation	10
3. Descriptive statistics	15
4. Graphical/distributional analysis	15
5. Variance and heterogeneity	15
6. Confidence intervals	10
7. Hypothesis testing	15
8. Covariance and correlation	10
9. Interpretation, confounding and conclusion	5
TOTAL	100

11. Data Source and Course References

- World Bank, World Development Indicators (WDI): https://datacatalog.worldbank.org/search/dataset/0037712/world-development-indicators
- World Bank Indicators API documentation: https://datahelpdesk.worldbank.org/knowledgebase/articles/889392
- World Bank Indicator API queries and metadata: https://datahelpdesk.worldbank.org/knowledgebase/articles/898599

Course material: Statistical Inference, Santosh Kumar Sahu, IIT Madras. The course material frames statistics as the collection, analysis, interpretation, and presentation of data; it covers secondary data, data management, descriptive statistics, relational statistics, confidence intervals, and hypothesis testing.

Course material: Confounding Variables, Santosh Kumar Sahu, IIT Madras. The material distinguishes correlation from causation and defines a confounder as a variable associated with the exposure and outcome that can create a spurious association.